

Project Development Phase

Date	8 July 2026
Team ID	
Project Name	Food Ordering And Customer Trends
Maximum Marks	4 Marks

1. Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the **Food Ordering Behaviour and Consumer Trends** dataset into Tableau Public.
- Verified that all 50,000 records and operational fields (Count of food_orders, Age Group, Cuisine, Meal Type, City, Order Value, Delivery Fee, Avg. Rating, Restaurant Type, Company Type, Mood, etc.) were imported successfully.
- Confirmed that the database structure was ready for active preprocessing.

Step 2: Check for Missing Values

- Examined the entire dataset for null, blank, or missing values across all structural attributes.
- Verified that critical transactional metrics such as Order Value, Delivery Fee, Customer Rating, Cuisine, Meal Type, and City contained fully populated, valid information.
- Handled any anomalies appropriately to prevent downstream errors during visual analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate order records using unique identifier metrics.
- Removed duplicate entries to prevent artificial inflation of metrics and inaccurate counts.
- Ensured that every distinct transaction was represented exactly once, preserving the true scale of the data (50,000 distinct records).

Step 4: Validate Data Types

Ensured that each attribute within the cleaned dataset was mapped to its correct structural data type for optimized filtering and visualization:

- **Count of food_orders** → Number (Integer)
- **Order Value** → Number (Decimal/Currency)
- **Delivery Fee** → Number (Decimal/Currency)
- **Avg. Rating** → Number (Decimal)
- **Avg. Time Taken To Order** → Number (Continuous)
- **Customer Age Group** → Categorical
- **Cuisine** → Categorical
- **Meal Type** → Categorical
- **Restaurant Type** → Categorical
- **Company Type** → Categorical
- **Mood** → Categorical
- **City** → Geographic / String

Step 5: Verify Data Consistency

- Scanned all rows for unrealistic or invalid values (such as negative order values, out-of-bounds ratings, or empty string errors).
- Standardized text inputs across categorical variables, ensuring consistent naming

formatting for cities, specific cuisines, and meal types.

- Verified logic boundaries across structural dimensions before proceeding to building the final charts.

Step 6: Prepare Calculated Fields & Aggregations

Created custom calculations, operational metrics, and explicit performance definitions required for the dashboard execution:

- Count of food_orders (Total dynamic order tracking)
- Total Delivery Fee (Summed logistical costs)
- Total Order Value (Gross transactional volume)
- Avg. Rating (Overall customer satisfaction level)
- Avg. Time Taken To Order (Mood-based time metrics)

Step 7: Final Clean Dataset Summary

- Completed validation checks to ensure the preprocessed dataset was accurate, structurally integrated, and completely ready for dashboard implementation.
- The finalized operational dataset was successfully loaded directly into the primary sheets of the Tableau Public workspace environment.

2. Business Questions with Implemented Visualization Matrix

S.No.	Business Question	Implemented Visualization
1	What are the total summary values, total volumes, logistical distribution fees, satisfaction averages after data preprocessing?	KPI Demonstration Matrix (Sheet 1) (Summary Card Table displaying 5 key metrics: 1.2M Total Orders, 2.98M Delivery Fees, 27.39M Order Value, 4.5K Avg. Rating, and 120K Avg. Time Taken To Order)
2	How do specific cuisine types compare in popularity when analyzed across different geographical distribution hubs?	City Vs Cuisine (Cross-tabulated Heatmap showing order distribution densities across 5 cities and 5 cuisines)
3	How do ordering habits split across different customer age categories based on the popularity of the restaurant segment?	Orders By Age Group & Restaurant Segment (Segmented Stacked Bar Chart mapping order volume by Age Group: Millennial, Gen Z, Adults, and Children)
4	What is the total proportional breakdown of the volume distribution of consumer preferences across core food groups?	Cuisine Preference Distribution (Color-coded Pie Chart mapping 5 cuisines: South Indian, Chinese, Italian, Biryani, Fast Food, and Desserts)

S.No.	Business Question	Implemented Visualization
5	What specific volume interaction pattern emerges when mapping social dining contexts across different meal periods?	Company Type & Meal Type Order Volume <i>(Grouped Multi-Variable Bar Chart comparing volume across Company Types: Solo, Alone, Family, Friends, and Partnerships)</i>
6	How are customer satisfaction ratings correlated with specific meal types across the major operational meal periods of the day?	Meal Type Vs Customer Rating <i>(Centered Donut Chart analyzing satisfaction levels across meal types)</i>
7	Which specific metropolitan regions show the single highest baseline of consumer order volume and transaction frequency?	Top 5 Cities By Order Volume <i>(Rank-sorted Descending Bar Chart showing volume for Mumbai, Pune, Bangalore, Delhi, and Chandigarh)</i>
8	How do daily dining financial values correlate directly with customer emotional state and ordering process durations?	Revenue Contribution By Meal Type & Ordering Based On Mood <i>(Dual-View Combo Visual featuring a Revenue Bar Chart and a Mood-based Average Duration Horizontal Bar Chart)</i>

Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

City Vs Cuisine

City	Cuisine				
	Biryani	Chinese	Desserts	Fast Food	North Indian
Bangalore	1,356	1,415	1,434	1,446	1,375
Chandigarh	1,374	1,408	1,378	1,396	1,351
Delhi	1,430	1,298	1,414	1,359	1,439
Hyderabad	1,377	1,389	1,357	1,369	1,380
Mumbai	1,397	1,397	1,463	1,409	1,395
Pune	1,366	1,444	1,406	1,389	1,358

Orders By Age Group

Age Group	Breakfast	Dinner	Lunch	Snacks
Millennial	2,574,716	2,524,296	2,491,205	2,552,133
Gen Z	2,639,355	2,442,804	2,500,027	2,558,469
Adults				
Child				

Order Preference Distribution

Preference	Count
Biryani	16.60%
Chinese	16.70%
Desserts	16.90%
Fast Food	16.74%
North Indian	16.46%
South Indian	16.60%

Company Type & Meal Type

Company of Order Id	Company / Meal Type				
	Alone	Family	Friends	Partner	
Breakfast	287	276	276	32	295
Dinner	276	276	276	32	295
Lunch	276	276	276	32	295
Snacks	276	276	276	32	295

Meal Type Vs Customer Rating

Meal Type	Rating
Breakfast	3.7
Dinner	3.8
Lunch	3.9
Snacks	3.5

Top 5 Cities By Order Volume

City	Order Volume
Mumbai	8,443
Pune	8,354
Bangalore	8,353
Delhi	8,309
Chandigarh	8,279

Sheet 1

Count of food_order...	Delivery Fee	Order Value	Avg. Rating...
50,000	2.98M	27.39M	3

Revenue Contribution By Meal Type

Meal Type	Revenue
Breakfast	6,924,753
Dinner	6,746,233
Lunch	6,855,039
Snacks	6,860,530

Ordering Based On Mood

Mood	Avg. Time Taken To Order
Lazy	~1.5
Happy	~1.5
Stressed	~1.5
Celebrating	~1.5

Story 1

